

A stand-alone proof of the even hybrid centre theorem for GL_1

This note proves the three clauses of **goalv0a**. It is the notation-migrated edition of August 10–11, 2026 of the refereed proof of July 19, 2026. The argument is unchanged — only symbols were renamed, by the Sandbox-wide dictionary $Q \mapsto q$, $q \mapsto v$, $z \mapsto t$: the relation formerly written $Q = q^2$ now reads $q = v^2$, and the family parameter formerly written z is now written t . (In the current Sandbox-wide frame, v and t denote the square roots $v^2 = q$ and $z^2 = t$; only v and t occur below.) Since September 15, 2026 the rings are printed in blackboard face, $\mathbb{A}_t$ and $\mathbb{F}_t$ — a change of face only: the two macro definitions and nothing else (revised [2026-09-15-12:08 · hypervisorA · Opus 5 Max · castalia:SandboxHQ]). The result identifies the even hybrid integral form of the family quantum GL_1 with its centre and with a Newton (divided-class) lattice; clause (iii) is the bilateral quantum Pólya theorem of Harman and Hopkins, transported to the multiplicative grid of the toral algebra (attribution remark at the end).

1 Conventions

Throughout, v and t are independent indeterminates and

$$\mathbb{A}_t = \mathbb{Z}[v^{\pm 1}, t^{\pm 1}], \quad \mathbb{F}_t = \mathbb{Q}(v)[t^{\pm 1}], \quad q = v^2.$$

Thus $\mathbb{A}_t \subseteq \mathbb{F}_t$ is a subring, and $\mathbb{Z}[q^{\pm 1}] = \mathbb{Z}[v^{\pm 2}] \subseteq \mathbb{Z}[v^{\pm 1}] \subseteq \mathbb{A}_t$.

The algebra U . U is the $\mathbb{F}_t$ -algebra with invertible pairwise-commuting generators $K_{n,+}, K_{n,-}$ ($n \in \mathbb{Z}$) subject to $K_{0,\pm} = 1$, $K_{m+n,\pm} = K_{m,\pm} K_{n,\pm}$ and $K_{n,+} K_{n,-} = t^n$. Writing $T^{\pm} = K_{1,\pm}$, the monomials $(T^+)^w$ ($w \in \mathbb{Z}$) form a $\mathbb{F}_t$ -basis; U is commutative. A *toral* element is an element of $\mathbb{F}_t[(T^+)^{\pm 1}]$; it is *even* when it is a Laurent polynomial in

$$\check{Y} := (T^+)^2.$$

Even toral elements thus form the Laurent-polynomial algebra $\mathbb{F}_t[\check{Y}^{\pm 1}]$.

Evaluations. For $\chi \in \mathbb{Z}$, let $\text{ev}_{\chi}: \mathbb{F}_t[\check{Y}^{\pm 1}] \rightarrow \mathbb{F}_t$ be the $\mathbb{F}_t$ -algebra homomorphism substituting the unit $q^{\chi} \in \mathbb{A}_t$ for $\check{Y}$, i.e.

$$\text{ev}_{\chi}\left(\sum_u c_u \check{Y}^u\right) = \sum_u c_u q^{u\chi} \quad (c_u \in \mathbb{F}_t).$$

It is the unique $\mathbb{A}_t$ -linear ring homomorphism with $\text{ev}_{\chi}(\check{Y}) = q^{\chi}$, and it fixes $\mathbb{F}_t$ (hence $\mathbb{A}_t$) pointwise. The *even Newton lattice* is

$$\mathcal{N}^{\text{ev}} = \{f \in \mathbb{F}_t[\check{Y}^{\pm 1}] : \text{ev}_{\chi}(f) \in \mathbb{A}_t \text{ for every } \chi \in \mathbb{Z}\}.$$

Because $\Phi^+ = \emptyset$ at GL_1 , the *even hybrid family integral form* is $U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}} \subseteq U$, with central part $Z(U^{\text{hyb, ev}}) = U^{\text{hyb, ev}} \cap Z(U \otimes_{\mathbb{F}_t} \text{Frac}(\mathbb{A}_t))$. Every element of U is toral, so the Harish–Chandra projection HC is the identity map on U ; the Weyl group is $W = \{1\}$, acting trivially, so $(\mathcal{N}^{\text{ev}})^W = \mathcal{N}^{\text{ev}}$.

Newton divided classes. For $r \geq 0$,

$$\nu_r(X) = \prod_{s=0}^{r-1} \frac{X - q^s}{q^r - q^s} \in \mathbb{Q}(v)[X], \quad \nu_0(X) = 1.$$

Each ν_r has degree exactly r (its leading coefficient $\prod_{s=0}^{r-1} (q^r - q^s)^{-1}$ is a nonzero element of $\mathbb{Q}(v)$), so $\{\nu_0, \dots, \nu_d\}$ is a $\mathbb{F}_t$ -basis of the module $\mathbb{F}_t[\check{Y}]_{\leq d}$ of even toral polynomials of degree $\leq d$.

We write $\mathbb{Z}[q^{\pm 1}]$ -integral for “lies in $\mathbb{Z}[q^{\pm 1}]$ ”. For $m \in \mathbb{Z}$ and $r \geq 0$ put

$$B_r(m) := \nu_r(q^m) = \prod_{s=0}^{r-1} \frac{q^m - q^s}{q^r - q^s}, \quad B_0(m) = 1.$$

2 Clause (i): $U^{\text{hyb, ev}}$ is an $\mathbb{A}_t$ -subalgebra

Proposition 1. $U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}}$ is an $\mathbb{A}_t$ -subalgebra of U . (This is the bilateral, t -spectator analogue of the fact that quantum integer-valued polynomials form a ring [1, §1].)

Proof. For each $\chi \in \mathbb{Z}$ the map $\text{ev}_\chi: \mathbb{F}_t[\check{Y}^{\pm 1}] \rightarrow \mathbb{F}_t$ is a ring homomorphism, and $\mathbb{A}_t \subseteq \mathbb{F}_t$ is a subring; hence the preimage $\text{ev}_\chi^{-1}(\mathbb{A}_t)$ is a subring of $\mathbb{F}_t[\check{Y}^{\pm 1}]$. It contains $\mathbb{A}_t$, because ev_χ fixes $\mathbb{A}_t$ pointwise, so $\text{ev}_\chi^{-1}(\mathbb{A}_t)$ is an $\mathbb{A}_t$ -subalgebra. By definition

$$\mathcal{N}^{\text{ev}} = \bigcap_{\chi \in \mathbb{Z}} \text{ev}_\chi^{-1}(\mathbb{A}_t)$$

is an intersection of $\mathbb{A}_t$ -subalgebras of $\mathbb{F}_t[\check{Y}^{\pm 1}]$, hence itself an $\mathbb{A}_t$ -subalgebra (it contains 1 since $\text{ev}_\chi(1) = 1 \in \mathbb{A}_t$, and is closed under sums, products and $\mathbb{A}_t$ -scalars). As $\mathbb{F}_t[\check{Y}^{\pm 1}] \subseteq U$, this is an $\mathbb{A}_t$ -subalgebra of U . $\square$

3 Clause (ii): the Harish–Chandra isomorphism is the identity

Proposition 2. The Harish–Chandra projection restricts to an isomorphism of $\mathbb{A}_t$ -algebras $\text{HC}: Z(U^{\text{hyb, ev}}) \xrightarrow{\sim} (\mathcal{N}^{\text{ev}})^W$, and this isomorphism is the identity map of $\mathcal{N}^{\text{ev}}$. (The rank-one, $\Phi^+ = \emptyset$ degeneration of the even hybrid Harish–Chandra isomorphism established at GL_2 in [2]; the classical ancestor is the Harish–Chandra isomorphism itself, whose quantum-group form the GL_2 record cites.)

Proof. The algebra U is commutative, being spanned by monomials in the pairwise commuting invertible generators $K_{n, \pm}$. Base change is compatible with commutativity, so $U \otimes_{\mathbb{F}_t} \text{Frac}(\mathbb{A}_t)$ is a commutative ring and equals its own centre. Therefore

$$Z(U^{\text{hyb, ev}}) = U^{\text{hyb, ev}} \cap Z(U \otimes_{\mathbb{F}_t} \text{Frac}(\mathbb{A}_t)) = U^{\text{hyb, ev}} \cap (U \otimes_{\mathbb{F}_t} \text{Frac}(\mathbb{A}_t)) = U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}},$$

the middle equality because $U \hookrightarrow U \otimes_{\mathbb{F}_t} \text{Frac}(\mathbb{A}_t)$ is injective — $\mathbb{F}_t \hookrightarrow \text{Frac}(\mathbb{A}_t) = \mathbb{Q}(v, t)$ and U is free over $\mathbb{F}_t$ — so $U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}} \subseteq U$ is literally a subset of the base-changed algebra. Every element of U is toral, so $\text{HC} = \text{id}_U$, and its restriction to $Z(U^{\text{hyb, ev}}) = \mathcal{N}^{\text{ev}}$ is $\text{id}_{\mathcal{N}^{\text{ev}}}$. Finally $W = \{1\}$ acts trivially, so $(\mathcal{N}^{\text{ev}})^W = \mathcal{N}^{\text{ev}}$. Hence

$$\text{HC}: Z(U^{\text{hyb, ev}}) = \mathcal{N}^{\text{ev}} \xrightarrow{\text{id}} \mathcal{N}^{\text{ev}} = (\mathcal{N}^{\text{ev}})^W$$

is the identity map of the $\mathbb{A}_t$ -algebra $\mathcal{N}^{\text{ev}}$ (an $\mathbb{A}_t$ -algebra by clause (i)), which is manifestly an isomorphism of $\mathbb{A}_t$ -algebras. $\square$

There is no content here beyond commutativity: at GL_1 the rational centre is everything, HC is the identity, and Weyl invariance is vacuous.

4 Clause (iii): the Newton presentation

Write

$$S := \sum_{u \in \mathbb{Z}} \sum_{r \geq 0} \mathbb{A}_t \check{Y}^u \nu_r(\check{Y}) \subseteq \mathbb{F}_t[\check{Y}^{\pm 1}]$$

for the $\mathbb{A}_t$ -submodule generated by the shifted divided classes $\check{Y}^u \nu_r(\check{Y})$.

Theorem 1 (Even Newton presentation for GL_1). $\mathcal{N}^{\text{ev}} = S = \sum_{u \in \mathbb{Z}} \sum_{r \geq 0} \mathbb{A}_t \check{Y}^u \nu_r(\check{Y})$. (The bilateral quantum Pólya theorem of Harman–Hopkins [1, §§1,4], transported to the multiplicative grid; the precise dictionary is the Attribution remark below.)

The proof occupies the rest of this section: the grid lemma, the forward inclusion $S \subseteq \mathcal{N}^{\text{ev}}$, the windowed reverse inclusion for polynomials, and the Laurent-clearing step.

4.1 The grid values are $\mathbb{Z}[q^{\pm 1}]$ -integral

Lemma 1. For all $m \in \mathbb{Z}$ and $r \geq 0$ one has $B_r(m) = \nu_r(q^m) \in \mathbb{Z}[q^{\pm 1}]$ (the Gaussian-binomial value pattern of [1, §1], extended to negative nodes as their §4 bilateral localization requires). Explicitly:

- (a) if $0 \leq m < r$ then $B_r(m) = 0$;
- (b) if $m \geq r \geq 0$ then $B_r(m) = \begin{bmatrix} m \\ r \end{bmatrix}_q \in \mathbb{Z}[q]$;
- (c) if $m = -n$ with $n \geq 1$ then

$$B_r(-n) = (-1)^r q^{-nr} \binom{r}{2} \begin{bmatrix} n+r-1 \\ r \end{bmatrix}_q \in \mathbb{Z}[q^{\pm 1}].$$

Proof. (a) If $0 \leq m < r$, the factor of the product at $s = m$ has numerator $q^m - q^m = 0$, so $B_r(m) = 0$.

(b) Assume $m \geq r \geq 0$. Pulling q^s out of each numerator and denominator factor,

$$B_r(m) = \prod_{s=0}^{r-1} \frac{q^s(q^{m-s} - 1)}{q^s(q^{r-s} - 1)} = \prod_{s=0}^{r-1} \frac{q^{m-s} - 1}{q^{r-s} - 1}.$$

Re-indexing by $i = r - s$ (so i runs from 1 to r),

$$B_r(m) = \prod_{i=1}^r \frac{q^{(m-r)+i} - 1}{q^i - 1} = \begin{bmatrix} m \\ r \end{bmatrix}_q,$$

the Gaussian binomial coefficient. It lies in $\mathbb{Z}[q]$: this is standard from the q -Pascal recurrence $\begin{bmatrix} m \\ r \end{bmatrix}_q = \begin{bmatrix} m-1 \\ r-1 \end{bmatrix}_q + q^r \begin{bmatrix} m-1 \\ r \end{bmatrix}_q$ with $\begin{bmatrix} m \\ 0 \end{bmatrix}_q = 1$, by induction on m .

(c) Let $m = -n$, $n \geq 1$. In each factor write the numerator as $q^{-n} - q^s = -q^{-n}(q^{n+s} - 1)$ and the denominator as $q^r - q^s = q^s(q^{r-s} - 1)$. Then

$$B_r(-n) = \prod_{s=0}^{r-1} \frac{-q^{-n}(q^{n+s} - 1)}{q^s(q^{r-s} - 1)} = (-1)^r q^{-nr} q^{-\sum_{s=0}^{r-1} s} \prod_{s=0}^{r-1} \frac{q^{n+s} - 1}{q^{r-s} - 1}.$$

Here $\sum_{s=0}^{r-1} s = \binom{r}{2}$. For the remaining product, the denominators give $\prod_{s=0}^{r-1} (q^{r-s} - 1) = \prod_{i=1}^r (q^i - 1)$ and the numerators give $\prod_{s=0}^{r-1} (q^{n+s} - 1) = \prod_{j=n}^{n+r-1} (q^j - 1)$, so

$$\prod_{s=0}^{r-1} \frac{q^{n+s} - 1}{q^{r-s} - 1} = \frac{\prod_{j=n}^{n+r-1} (q^j - 1)}{\prod_{i=1}^r (q^i - 1)} = \frac{[n+r-1]_q! / [n-1]_q!}{[r]_q!} = \begin{bmatrix} n+r-1 \\ r \end{bmatrix}_q,$$

where $[k]_q = (q^k - 1)/(q - 1)$ and the common powers of $(q - 1)$ cancel. The Gaussian binomial $\begin{bmatrix} n+r-1 \\ r \end{bmatrix}_q \in \mathbb{Z}[q]$ by part (b). Collecting the prefactor gives

$$B_r(-n) = (-1)^r q^{-nr - \binom{r}{2}} \begin{bmatrix} n+r-1 \\ r \end{bmatrix}_q \in \mathbb{Z}[q^{\pm 1}]. \quad \square$$

4.2 Forward inclusion $S \subseteq \mathcal{N}^{\text{ev}}$

Lemma 2. *For every $u \in \mathbb{Z}$ and $r \geq 0$, $\check{Y}^u \nu_r(\check{Y}) \in \mathcal{N}^{\text{ev}}$. Hence $S \subseteq \mathcal{N}^{\text{ev}}$. (The transported forward half of [1, §1]: q -binomials are quantum integer-valued.)*

Proof. Fix u, r and $\chi \in \mathbb{Z}$. Since ev_χ is a ring homomorphism with $\text{ev}_\chi(\check{Y}) = q^\chi$,

$$\text{ev}_\chi(\check{Y}^u \nu_r(\check{Y})) = q^{u\chi} \nu_r(q^\chi) = q^{u\chi} B_r(\chi).$$

By Lemma 1, $B_r(\chi) \in \mathbb{Z}[q^{\pm 1}] \subseteq \mathbb{A}_t$, and $q^{u\chi}$ is a unit of $\mathbb{A}_t$; so the product lies in $\mathbb{A}_t$. As χ was arbitrary, $\check{Y}^u \nu_r(\check{Y}) \in \mathcal{N}^{\text{ev}}$. The lattice $\mathcal{N}^{\text{ev}}$ is an $\mathbb{A}_t$ -module (indeed an $\mathbb{A}_t$ -algebra, clause (i)), so it contains every $\mathbb{A}_t$ -combination of these generators, i.e. $S \subseteq \mathcal{N}^{\text{ev}}$. $\square$

4.3 Reverse inclusion for polynomials: windowed Newton interpolation

Lemma 3. *Let $g \in \mathcal{N}^{\text{ev}}$ be an even toral polynomial, $g \in \mathbb{F}_t[\check{Y}]$, of degree $\leq d$. Then $g = \sum_{r=0}^d c_r \nu_r(\check{Y})$ with all $c_r \in \mathbb{A}_t$. In particular $g \in \sum_{r \geq 0} \mathbb{A}_t \nu_r(\check{Y}) \subseteq S$. (Pólya's windowed-interpolation argument as q -deformed in the proof of [1, §1], run on the multiplicative grid with t a spectator.)*

Proof. Because $\{\nu_0, \dots, \nu_d\}$ is a $\mathbb{F}_t$ -basis of $\mathbb{F}_t[\check{Y}]_{\leq d}$, there are unique $c_0, \dots, c_d \in \mathbb{F}_t$ with

$$g = \sum_{r=0}^d c_r \nu_r(\check{Y}).$$

Evaluate on the window $\chi = m$ for $m = 0, 1, \dots, d$. As ev_m is $\mathbb{F}_t$ -linear and $\text{ev}_m(\nu_r(\check{Y})) = B_r(m)$,

$$\text{ev}_m(g) = \sum_{r=0}^d B_r(m) c_r = \sum_{r=0}^d V[m, r] c_r, \quad V[m, r] := B_r(m) \quad (0 \leq m, r \leq d).$$

By Lemma 1, every entry $V[m, r] \in \mathbb{Z}[q^{\pm 1}]$; moreover V is *lower unitriangular*: $V[m, r] = B_r(m) = 0$ for $m < r$ (part (a)), and $V[m, m] = B_m(m) = \prod_{s=0}^{m-1} \frac{q^m - q^s}{q^m - q^s} = 1$. Hence $\det V = 1$, and the inverse V^{-1} is again lower unitriangular with entries in $\mathbb{Z}[q^{\pm 1}]$ (back-substitution never leaves $\mathbb{Z}[q^{\pm 1}]$, and by uniqueness this inverse coincides with the inverse taken over the larger ring $\mathbb{F}_t$).

Writing $\mathbf{c} = (c_0, \dots, c_d)^\top$ and $\mathbf{v} = (\text{ev}_0(g), \dots, \text{ev}_d(g))^\top$, we have $\mathbf{v} = V\mathbf{c}$ in $\mathbb{F}_t^{d+1}$, hence

$$\mathbf{c} = V^{-1}\mathbf{v}.$$

Since $g \in \mathcal{N}^{\text{ev}}$, each $\text{ev}_m(g) \in \mathbb{A}_t$, so $\mathbf{v} \in \mathbb{A}_t^{d+1}$; and V^{-1} has entries in $\mathbb{Z}[q^{\pm 1}] \subseteq \mathbb{A}_t$. Therefore $\mathbf{c} = V^{-1}\mathbf{v} \in \mathbb{A}_t^{d+1}$, i.e. every $c_r \in \mathbb{A}_t$.

The t -spectator argument. The map ev_m substitutes only $\check{Y} \mapsto q^m = v^{2m}$; it does not touch v or t . Thus t never enters the matrix V , whose entries lie in $\mathbb{Z}[q^{\pm 1}]$. Concretely, expand each coordinate along the free $\mathbb{Z}[v^{\pm 1}]$ -basis of t -powers: $c_r = \sum_k c_{r,k} t^k$ with $c_{r,k} \in \mathbb{Q}(v)$, so that $\text{ev}_m(g) = \sum_k (\sum_r B_r(m) c_{r,k}) t^k$. Membership $\text{ev}_m(g) \in \mathbb{A}_t = \mathbb{Z}[v^{\pm 1}, t^{\pm 1}]$ means that for

each fixed k the t^k -coefficient $\sum_r B_r(m) c_{r,k}$ lies in $\mathbb{Z}[v^{\pm 1}]$ for all $m \in \{0, \dots, d\}$. For each fixed k this is precisely the unitriangular system above, over $\mathbb{Z}[q^{\pm 1}]$ with right-hand sides in $\mathbb{Z}[v^{\pm 1}]$; it gives $c_{r,k} \in \mathbb{Z}[v^{\pm 1}]$. Hence $c_r = \sum_k c_{r,k} t^k \in \mathbb{Z}[v^{\pm 1}, t^{\pm 1}] = \mathbb{A}_t$, as claimed. This is the same conclusion reached above; we spell out the coefficient-wise reading to make the spectator role of t explicit.

Base change $\mathbb{Z}[q^{\pm 1}] \rightarrow \mathbb{Z}[v^{\pm 1}] \rightarrow \mathbb{A}_t$. Two rings appear: the matrix V and its inverse live over $\mathbb{Z}[q^{\pm 1}] = \mathbb{Z}[v^{\pm 2}]$, whereas the coordinates c_r and the values $\text{ev}_m(g)$ live over $\mathbb{F}_t = \mathbb{Q}(v)[t^{\pm 1}]$ — honest v (odd powers) may occur, so the c_r need not lie in $\mathbb{Q}(q)$. This causes no loss: the extension $\mathbb{Z}[q^{\pm 1}] \hookrightarrow \mathbb{Z}[v^{\pm 1}]$ is free with basis $\{1, v\}$, hence flat, and multiplication by a $\mathbb{Z}[q^{\pm 1}]$ -entry preserves each of the two grades (even/odd powers of v). The reverse computation $\mathbf{c} = V^{-1}\mathbf{v}$ therefore runs identically in each grade, so integrality of the values over $\mathbb{Z}[v^{\pm 1}, t^{\pm 1}]$ forces integrality of the coordinates over the same ring. Equivalently, and most directly: V^{-1} has entries in $\mathbb{Z}[q^{\pm 1}] \subseteq \mathbb{A}_t$ and $\mathbf{v} \in \mathbb{A}_t^{d+1}$, so $\mathbf{c} = V^{-1}\mathbf{v} \in \mathbb{A}_t^{d+1}$. $\square$

4.4 Laurent clearing and conclusion

Proof of Theorem 1. The inclusion $S \subseteq \mathcal{N}^{\text{ev}}$ is Lemma 2. For $\mathcal{N}^{\text{ev}} \subseteq S$, let $f \in \mathcal{N}^{\text{ev}} \subseteq \mathbb{F}_t[\check{Y}^{\pm 1}]$ be arbitrary. Choose $N \geq 0$ so large that $g := \check{Y}^N f \in \mathbb{F}_t[\check{Y}]$ is an honest polynomial (take N at least the largest magnitude of a negative $\check{Y}$ -exponent occurring in f). For every $\chi \in \mathbb{Z}$, using that ev_χ is a ring homomorphism,

$$\text{ev}_\chi(g) = \text{ev}_\chi(\check{Y}^N) \text{ev}_\chi(f) = q^{N\chi} \text{ev}_\chi(f).$$

Here $q^{N\chi}$ is a unit of $\mathbb{A}_t$ and $\text{ev}_\chi(f) \in \mathbb{A}_t$, so $\text{ev}_\chi(g) \in \mathbb{A}_t$; thus $g \in \mathcal{N}^{\text{ev}} \cap \mathbb{F}_t[\check{Y}]$. By Lemma 3, $g = \sum_r c_r \nu_r(\check{Y})$ with $c_r \in \mathbb{A}_t$. Multiplying back by the unit $\check{Y}^{-N}$,

$$f = \check{Y}^{-N} g = \sum_{r \geq 0} c_r \check{Y}^{-N} \nu_r(\check{Y}) \in \sum_{u \in \mathbb{Z}} \sum_{r \geq 0} \mathbb{A}_t \check{Y}^u \nu_r(\check{Y}) = S,$$

each summand being an $\mathbb{A}_t$ -multiple of the generator $\check{Y}^u \nu_r(\check{Y})$ with $u = -N$. Hence $\mathcal{N}^{\text{ev}} \subseteq S$, and with Lemma 2, $\mathcal{N}^{\text{ev}} = S$. $\square$

5 Attribution

Remark 1. *Clause (iii) is the quantum Pólya theorem of Harman and Hopkins, Quantum integer-valued polynomials (arXiv:1601.06110), in bilateral Laurent form. Their §1 Proposition (Prop. §1.2, **prop:qbinom**), the q -analogue of Pólya's 1919 theorem, states that the ring $\mathcal{R}_q^+ = \{P \in \mathbb{Q}(q)[x] : P([n]_q) \in \mathbb{Z}[q] \ \forall n \in \mathbb{N}\}$ is freely generated as a $\mathbb{Z}[q]$ -module by the*

q -binomial coefficient polynomials $\begin{bmatrix} x \\ k \end{bmatrix}_q$ (the q -deformed Newton basis on the additive grid

*$x = [n]_q$, with denominator $q^{\binom{k}{2}} [k]_q!$ — not the classical Pólya $\binom{x}{k}$); their §4 Proposition (Prop. §4.3, **prop:localization**), $\mathcal{R}_q = \mathcal{R}_q^+ \otimes_{\mathbb{Z}[q]} \mathbb{Z}[q^{\pm 1}]$, is the bilateral localization. In the current notation their parameter q coincides with our q ($= v^2$), so their displayed formulas read verbatim on our side.*

The node-preserving affine substitution $\check{Y} = 1 + (q - 1)x$ is exactly their §4 z -variable; it carries the grid node $x = [n]_q$ to $\check{Y} = q^n$ and carries each basis term $\begin{bmatrix} x \\ k \end{bmatrix}_q$ onto $\nu_k(\check{Y})$ term by term — the factors $(q - 1)$ cancel in every Newton ratio. (They remark further that on adjoining square roots $K^2 = z$, $v^2 = q$ — which is precisely the passage to our T^+ and our v — the ring $\mathcal{R}_q[K, v]$ is equivalent to the Cartan part of Lusztig's integral form of $U_v(\mathfrak{sl}_2)$).

Nothing below rests on that remark.) The values-in- $\mathbb{Z}[q^{\pm 1}]$ side matches values-in- $\mathbb{A}_t$ with t a spectator, since evaluation is t -coefficient-wise and $\mathbb{Q}(q) \cap \mathbb{Z}[v^{\pm 1}] = \mathbb{Z}[q^{\pm 1}]$.

The two-variable (GL_2) form of this equivalence is established in [2]: there the grid criterion agrees with membership in the $\mathbb{A}_t$ -span of the product Newton generators, and passes to the full Laurent even lattice. The GL_1 statement above is the rank-one specialization of that transport.

References

- [1] N. Harman and S. Hopkins, *Quantum integer-valued polynomials*, arXiv:1601.06110 (2016).
- [2] SandboxA, *The centre of the even hybrid family quantum GL_2* (closure record of the signed goal, revision A1+R5), Sandbox record, 2026.